

Probing Timbre Space: Gate Experiments for Inverse Synthesizer Patch Synthesis

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Abstract

A system that turns human intent—a text description or a hummed melody—into a *structured synthesizer parameter set* (a patch) rather than audio must clear two assumptions before it is worth building: (i) an embedding distance between two sounds has to track human similarity judgements, and (ii) that distance has to be usable as an objective function for search over patch parameters. We treat these as *gate hypotheses*, build a testbed that can falsify them at low cost, and report what it says.

The testbed (*timbreprobe*) consists of a differentiable subtractive synthesizer (31 parameters, STFT-domain time-varying filtering, bit-deterministic renders), an 85-dimensional differentiable DSP embedding, an anchor-stratified corpus of 3,895 patches of which four entire anchor regions are held out, a learned patch $\rightarrow$ embedding surrogate, and seven search strategies that all start from the same retrieval-based initialisation per target.

On the navigability gate the answer is *weakly positive*, and the way it fails is the informative part. True-gradient search is the only method that beats its retrieval start (median distance ratio 0.875, 95% CI [0.81, 0.92]), but it plateaus after roughly 15% and consumes 5 856 real renders. A derivative-free baseline (separable CMA-ES) spends $3.7\times$ more renders and makes no measurable progress at all (1.00: its median target is unchanged). All three surrogate-based strategies end *worse than not searching* (ratio 1.36–2.59), and—the diagnostic signature—they degrade *more* in-distribution than on held-out targets, i.e. the optimiser exploits surrogate error precisely where the surrogate is most confident. Surrogate distance-rank fidelity falls from Kendall $\tau = 0.70$ in-distribution to 0.39 on unseen regions, while the true-gradient oracle is unaffected. The bottleneck is therefore the surrogate, not the search algorithm; the dominant failure mode is extrapolation plus exploitation rather than interpolation error.

We release the testbed, the pre-registered listening-test material for the empirical gate, and a roadmap in which every subsequent stage has an explicit gate and a falsification criterion. Six engineering findings that cost real time are documented—a per-sample IIR loop is $100\times$ slower than an equivalent spectral formulation, multi-threaded torch can be $2\text{--}5\times$ slower than single-threaded at these tensor shapes, `torch.istft` backward produced non-finite gradients at padded edges, an `asin`-based triangle LFO has an infinite derivative at its extrema, fixed-step Adam overshoots by $1.7\times$ in distance within five steps, and a corpus jitter layer can place test patches within numerical noise of training patches, which makes the improvement ratio degenerate unless targets are filtered.

1 Introduction

The problem. Building a sound is a search problem. A producer who wants “a warm detuned pad with a slow filter opening” does not know which of a modern synthesizer’s several hundred

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parameters to move; they iterate, listening between attempts. Recent products automate parts of that loop. Fadr’s *SynthGPT 2* accepts a text description and returns a playable patch, with a human listening between candidates; Sonic Charge’s *Genopatch* evolves patches toward an imported audio sample [1, 2]. Both are impressive, and both are *closed-loop without a perceptual objective*: the machine proposes, a human decides, and nothing in the system knows *why* one candidate was preferred.

Our larger goal is a system that maps human intent—text, a humming reference, or both—to a patch, and that can improve from listening feedback rather than only from a human gate. Architecturally that requires two things before any generative machinery is justified.

Assumption A (perceptual monotonicity). There exists an embedding space of audio in which distance predicts human similarity judgements for timbre, at least ordinally. If not, “optimising toward a target embedding” is optimising toward nothing.

Assumption B (navigability). Given a target in that space, a search procedure can find patch parameters that reach it, at a render budget a user would tolerate. If not, the loop must be retrieval-and-rerank only.

This paper tests A and B *before* building the system, and reports what the tests revealed. We deliberately chose a testbed whose oracle is exact—a differentiable synthesizer—so that “true gradient” navigation is available as an upper bound and render budgets can be accounted for honestly. This turned what we expected to be a calibration exercise into a result: the naive design (learn a surrogate, optimise freely on it) fails, and it fails in a way that is measurable, diagnosable, and cheap to detect in advance.

Contributions.

- (1) A *differentiable timbre testbed*: a 31-parameter subtractive synthesizer whose time-varying filter is implemented in the STFT domain (validated against a per-sample state-variable reference at Spearman $\rho = 0.996$), with bit-deterministic renders and an 85-dimensional differentiable DSP embedding, giving an exact oracle for patch $\rightarrow$ percept gradient navigation at CPU-feasible cost (§3).
- (2) A *gate protocol* with three experiments: E2 (navigability, fully automatic), E3 (surrogate fidelity, reported per *region* rather than in aggregate), and E1 (a pre-registered triplet 2AFC listening test whose material the code generates; human data pending) (§4).
- (3) *Negative results with a signature*: all surrogate-based search degrades below the no-search baseline, and the degradation is largest where the surrogate is most confident—a measurement that costs one extra data split, and that we recommend as a standard diagnostic (§5.3).
- (4) A *render-efficiency comparison* showing that gradient information beats population search by more than $3\times$ in render budget at this dimensionality, which argues for investing in differentiable proxies rather than larger populations (§5.2).
- (5) *Diagnostics that localise the failure*: a residual decomposition, a discrete-jump correlation, an ablation of the monotone accept rule, and a block-balanced-metric control (§5.5). Together they rule out the accept rule as the binding constraint, show that the leftover error is spread across the objective roughly in proportion to dimensionality (per-dimension shares $0.85\text{--}2.34\times$ uniform, i.e. only mildly overweighted in the discrete-sensitive blocks), and motivate re-weighting or learning the metric rather than a more elaborate search.
- (6) A *documented corpus trap*: a jitter-sampled corpus places $\sim 4\%$ of test patches within numerical noise of training patches, making the improvement ratio unbounded for those targets; the fix (a scale-free minimum target distance) and its effect on reported means are described in §4.1.
- (7) An *evidence-linked roadmap* (§7) in which every next step is tied to the observation that motivates it and to a criterion that would falsify it.

2 Related work

2.1 Inverse synthesis and preset generation

Learning to map from sound to synthesizer control has a long history, from flow-based synthesis of parameters from audio [3] to differentiable digital signal processing, where a differentiable synthesizer is placed inside the training loop [4]. Percussive audio has been mapped to synthesizer parameters in real time with a differentiable drum synthesizer [5], and the non-proportional geometry between synth parameters and their perceptual effect has been studied directly [6]. Commercially, *Genopatch* evolves patches toward a target sample by genetic search in parameter space [2], and *SynthGPT 2* generates patches from text with a human in the loop [1].

Two things distinguish our work from this line. First, it does not propose a generator: it asks what an honest objective for such a generator must look like, and measures the cost of getting it wrong. Second, its evaluation protocol forces all methods to start from the same retrieval point, which changes the conclusions one draws about how much optimisation is worth (see §5.2 and §6.4).

2.2 Perceptual embeddings for timbre, and their validation

Wessel’s timbre space established that timbre is navigable in a low-dimensional perceptual geometry [7], but the coordinates were hand-designed. Modern audio–text and music representation models—CLAP [8], MERT [9], MuQ [10]—provide learned spaces in which distance is often *assumed* to be perceptually meaningful. Our E1 protocol is a way to test that assumption for timbre specifically, and our testbed deliberately keeps a handcrafted embedding as a lower anchor: a pretrained model that cannot beat handcrafted spectral features on human similarity judgements is not a perceptual model for this task, whatever its retrieval benchmarks say.

2.3 Surrogates, trust regions, and model exploitation

Optimising a learned model of an expensive objective is standard practice, and so is its central failure mode: the optimiser finds regions where the model is wrong [11]. Trust-region methods constrain the search to where the model is locally valid [11]; our E2 result quantifies the same phenomenon in a perceptual setting, shows that the damage is *larger* in-distribution than out-of-distribution (the signature of exploitation rather than mere generalisation error), and shows that a trust region recovers only part of the loss. For derivative-free comparison we use separable CMA-ES [12] within the CMA-ES family [13]; gradient search uses Adam [14].

2.4 Humming and intent

Humming as a query is attractive because it carries pitch and articulation directly. But its *source* is a human voice while the target is a synthesizer, so a naïve audio embedding answers “imitate this singer”—the wrong question. The roadmap therefore treats humming as an intent-factorisation problem rather than audio-to-audio matching: pitch trajectory → tuning parameters; posture (attack sharpness, vibrato rate and depth, legato) → envelopes and LFO; brightness and vowel colour → filter and drive (§7).

3 The *timbreprobe* testbed

3.1 Design constraints

A gate experiment needs its oracle to be exact, its cost to be accounted, and its conclusions to be attributable. We therefore build the synthesizer rather than adopting one:

- **Exact gradients.** Because the entire chain parameter $\rightarrow$ audio $\rightarrow$ embedding is differentiable, “true gradient” navigation is available as an upper bound that no surrogate can exceed. Without it, a failed surrogate experiment cannot be distinguished from a hard landscape.
- **Render accounting.** Every real render is counted, so methods can be compared by *quality per render* rather than per iteration—the resource a practical system actually spends.
- **Controlled difficulty.** A 31-parameter space is small enough that a corpus can cover it in layers and large enough to contain the phenomena of interest (coupled parameters, discrete choices, log-scaled axes).
- **Determinism.** Renders are bit-identical for identical patches (fixed noise seed), which the search methods rely on.

The cost of this choice is realism: our synthesizer is not a specific commercial instrument, has no effects section, and filters without phase response (§3.4). It is an oracle for gate hypotheses, not a substitute for a target instrument; §7 makes the port to a real instrument an explicit, gated phase.

3.2 Parameter schema and its representation

The synthesizer exposes $D = 31$ parameters (Table 6 in Appendix A). The optimiser always works in the unit hypercube $u \in [0, 1]^D$; physical units are recovered by

$$\text{linear: } v = \ell + u(h - \ell), \quad \text{log: } v = \ell(h/\ell)^u, \quad (1)$$

where ℓ, h are the physical bounds. Log scaling is used for the cutoff, Q , and the envelope times and LFO rate, because a factor of two in cutoff is a perceptually equal step everywhere while an additive step is not.

Discrete parameters (waveform, octave, filter type, LFO target and waveform, and the pulse-duty choice) are represented as a continuous *position* over the ordered choice list. With k choices and $p = u(k - 1)$,

$$w_j = \frac{\max(0, 1 - |p - j|)}{\sum_{j'} \max(0, 1 - |p - j'|)}, \quad \hat{\theta} = \sum_j w_j \theta_j, \quad (2)$$

i.e. a piecewise-linear morph between adjacent choices, and the *hardened* patch used for reporting is $j^* = \text{round}(p)$.

This choice of kernel, rather than a softmax or Gumbel relaxation, is deliberate. Real preset patches sit exactly *on* a choice, and a softmax kernel’s derivative vanishes there, which would freeze discrete dimensions for gradient search at the very points where the corpus is dense. The triangular basis satisfies $|\partial w_j / \partial p| = 1$ throughout and is a partition of unity on $[0, k - 1]$, so gradients stay alive everywhere and the parameter reads as an interpretable morph knob.

3.3 Signal chain

The chain is

$$\text{osc}_1(+\text{PM}) + \text{osc}_2 + \text{sub} + \text{noise} \rightarrow \text{drive} \rightarrow \text{filter} \rightarrow \text{VCA} \rightarrow \text{master} \rightarrow \text{RMS}, \quad (3)$$

rendered at $f_s = 16 \text{ kHz}$ for 0.25 s on a fixed C4 (261.63 Hz).

Oscillators. Each oscillator is additive over $K = 24$ harmonics with per-family unit-RMS coefficient vectors:

$$a_1^{\text{sine}} = 1; \quad a_k^{\text{saw}} = \frac{1}{k}; \quad a_k^{\text{tri}} = \frac{1}{k^2} \text{ (} k \text{ odd)}; \quad a_k^{\text{pulse}}(d) = \frac{2}{k\pi} \sin(\pi kd), \quad (4)$$

normalised so that $\sum_k a_k^2/2 = 1$. The families are blended with the weights of Eq. (2), and each harmonic is masked to prevent aliasing at the render pitch:

$$m_k = \sigma\left(\frac{0.95f_{\text{Nyq}} - kf_0(1 + 0.6\text{FM})}{0.05f_{\text{Nyq}}}\right), \quad o(t) = \sum_{k=1}^K m_k a_k \sin(k\phi(t)), \quad (5)$$

with phase obtained by integrating the instantaneous frequency and adding phase modulation,

$$\phi(t) = \frac{2\pi}{f_s} \sum_{\tau \leq t} f(\tau) + I o_2(t), \quad f(t) = f_{C4} 2^{\text{oct} + \text{cents}/1200} \cdot 2^{w_{\text{pitch}} D_{\text{LFO}} 7 \text{LFO}(t)/12}. \quad (6)$$

All harmonic phases are sine phases: the embedding is magnitude-based and thus phase-invariant, and the ear is largely phase-insensitive for steady tones, so this costs nothing here and keeps the synthesis differentiable. The sub oscillator is a sine one octave below oscillator 1 (i.e. at $f/2$); noise is a white/pink blend, the pink component shaped as $1/\sqrt{f}$ in the frequency domain from a fixed-seed white signal.

Modulation and envelopes. The LFO is a closed-form waveform set—sine, triangle $1 - 4|\text{frac}(\varphi + 0.25) - 0.5|$, softened square $\tanh(6 \sin 2\pi\varphi)$, saw $2\text{frac}(\varphi) - 1$ —blended with Eq. (2) and routed to cutoff (± 3 octaves), pitch (± 7 semitones) or amplitude (tremolo depth 0.5). Envelopes (filter and amplitude) are

$$\text{env}(t) = \begin{cases} t/a, & t < a, \\ s + (1-s)e^{-(t-a)/\tau_d}, & t \geq a, \end{cases} \quad \tau_d = d/5, \quad (7)$$

a linear attack into an exponential decay, continuous at $t = a$ and differentiable everywhere except that single point.

Filter, VCA, output. The filter is a time-varying state-variable filter, described in §3.4. The VCA multiplies by $\text{env}_a(t)$ and the tremolo, the output stage applies $\tanh(g_m \cdot y)$ with $g_m \in [1, 8]$, and the render is finally RMS-normalised to 0.1, so loudness is not a distance cue (mirroring loudness matching in listening tests).

3.4 Filtering in the STFT domain

A per-sample zero-delay-feedback state-variable filter [15] was the first implementation, and it was unusable: **10 s per batch forward and 60 s with backward**, essentially independent of batch size, because 6400 Python-level iterations dominate. We therefore apply the *analytic magnitude response* of the same filter family once per STFT frame ($n_{\text{fft}} = 256$, hop 64, i.e. 4 ms control rate):

$$r = \frac{f}{f_c(\text{frame})}, \quad |H_{lp}| = [(1-r^2)^2 + (r/Q)^2]^{-1/2}, \quad |H_{hp}| = r^2 |H_{lp}|, \quad |H_{bp}| = \frac{r}{Q} |H_{lp}|, \quad (8)$$

with LP/HP/BP and 12/24 dB/oct as soft blends ($H = (1-\lambda)H_{12} + \lambda H_{12}^2$), and the per-frame cutoff

$$f_c(t) = \text{clamp}\left(f_{c0} 2^{A_e \text{env}_f(t)} 2^{w_{\text{cut}} D_{\text{LFO}} 3 \text{LFO}(t)}, 20, 0.92f_{\text{Nyq}}\right). \quad (9)$$

Synthesis uses a hand-rolled overlap-add with a clamped window-envelope normaliser (not `torch.istft`, whose backward divides by a near-zero envelope at the padded edges and produced non-finite gradients).

What this costs, and why the experiments survive it. The substitution preserves per-frame time-varying cutoff, the Q peak, and the type/slope blends. It removes the filter’s *phase* response: no ringing, no self-oscillation, and spectral transients are smoothed at the 16 ms analysis window. Three design decisions limit the consequences. (i) The amplitude envelope is applied *after* synthesis, so loudness transients—what the envelope features measure—stay sample-accurate. (ii) Every embedding dimension is a magnitude or an envelope statistic; none depends on phase. (iii) The engineering claim is validated directly: on a fixed test signal the spectral filter’s log-magnitude spectrum agrees with the per-sample reference at Spearman $\rho = 0.996$ (checked by `sanity`). What a real instrument port must re-check is exactly the part we removed—transient bite and resonance character—which is why §7 makes the port a separate gated phase rather than an assumption.

3.5 The embedding

The embedding is 85 handcrafted, differentiable dimensions, grouped into 13 blocks (Appendix B): 32 log-mel bands with mean and standard deviation (64 dims), spectral centroid, bandwidth, soft rolloff, flatness, flux and slope (mean/std each), crest factor, a high/low band ratio, four envelope statistics, a smooth zero-crossing proxy, and harmonic and odd/even harmonic ratios referenced to the patch’s fundamental. Features are z -scored with corpus statistics.

Two properties are engineered deliberately.

Smoothness. Every hard, non-differentiable operation is replaced by a smooth surrogate, because a feature with zero gradient almost everywhere contributes nothing to gradient navigation while still distorting distances: the rolloff threshold becomes a sigmoid weighting ($w_f = \sigma((\text{cum}_f - 0.85 \text{ tot})/(0.05 \text{ tot}))$), the envelope’s peak position becomes a log-sum-exp, and the zero-crossing rate becomes $\frac{f_s}{2} \langle \sigma(-x_t x_{t+1} / \sigma_x^2) \rangle$ with $\sigma_x^2 = 0.05 \langle x^2 \rangle$.

Harmonic structure. Two dimensions are computed against the patch’s own fundamental f_0 via Gaussian combs $w_f^{(k)} = \exp(-((f - k f_0)/(0.35 f_0))^2)$: the total energy captured by the comb (harmonicity) and the odd-to-even ratio (which separates square- from saw-family timbres). These are the dimensions most sensitive to *discrete* choices, and they matter in the diagnostics of §5.5.

The embedding plays two roles: it is the objective for E2, and it is the *lower anchor* for E1—a pretrained embedding that cannot beat it on human judgements is not worth adopting. It is not claimed to be perceptual.

3.6 Corpus construction

Uniform random sampling of a 31-dimensional synth space is a poor corpus: most of the volume is perceptually degenerate (nothing audible; everything wide open). The corpus is built in layers around 16 hand-designed anchors with semantic labels (bright lead, warm pad, dark sub, hollow square, narrow pulse, FM brass, noise breath, pluck, resonant sweep, soft flute, triangle organ, HP air, octave stack, sub-heavy pad, driven lead, metallic FM), of which *four are held out entirely* and serve as generalisation probes:

- *sweeps*: for every anchor and every dimension, three values moved across the range with the anchor fixed ($31 \times 3 \times 16$, deduplicated)—these map each axis’s *local* perceptual direction;
- *jitter*: 120 Gaussian perturbations per anchor ($\sigma = 0.10$ continuous, 0.03 choice, so patches stay close to “hard” presets)—these populate the neighbourhoods optimisation actually traverses;
- *uniform* (500) and *sparse* (150, one or two sources active) for global coverage.

Table 1: Identifiability under the render protocol.

not identifiable	why
envelope release stages	no note-off in the protocol
slow LFOs (below ~ 4 Hz)	less than one cycle in 0.25 s
absolute loudness	render is RMS-normalised
source level, if it is the only active source	level is absorbed by the normaliser
oscillator 2 amplitude under FM	FM index and level trade off

After deduplication on rounded coordinates this yields 3,895 unique patches, split into train 2,468 / val 308 / test 1,119 (of which 811 belong to the held-out anchor regions). The corpus is a scientific instrument: neutral and controlled. A Stage-1 retrieval library would instead use *real* factory presets, where a biased distribution is a feature, not a bug.

A corpus trap worth reporting. Because the jitter layer is dense, some test patches land within numerical noise of a training patch: in the smoke corpus, 43 of roughly one thousand test candidates have a nearest-training-neighbour distance below a quarter of the median, and a few are numerically *identical* to a training patch. Such targets are already solved by the retrieval start, so their “improvement” ratio $d_{\text{final}}/d_{\text{init}}$ is unbounded—left in, three such targets dragged the *mean* ratio of every method to $\sim 10^7$ while leaving medians untouched. E2 therefore rejects targets closer than $0.25\times$ the median nearest-neighbour distance and resamples (§4.1). Anyone building a sampled patch corpus should check for this before reporting ratio means.

3.7 Render protocol and identifiability

The protocol is a single 0.25 s sustained C4, no note-off. This makes the comparison clean (every patch is heard as the same note, so “timbre” is isolated from “melody”) and it makes the experiment cheap, at the price of identifiability. Table 1 states what is and is not recoverable; parameter-recovery numbers must be read against it.

4 Experimental protocol

4.1 E2: navigability

Targets and the shared start. All methods start each target from the *same* point: the nearest *training* patch in embedding space—the retrieval initialisation a practical system would use. 48 targets are sampled from the test split, deliberately balanced (24 from held-out anchor regions, the rest in-distribution) so that generalisation is measured rather than assumed, and filtered against near-duplicates as described in §3.6. Every reported number is therefore the value added *on top of retrieval*; a method that cannot beat its own start point has negative value.

Methods.

- `retrieval_nearest` — the start point itself (ratio 1.000 by construction);
- `retrieval_unigram64` — best of 64 random training presets, the “blind trial-and-error” floor;
- `grad_surrogate` — adaptive-step gradient search on the learned surrogate (200 steps, zero renders during search);
- `grad_surrogate_trust` — the same, confined to an L_2 ball of radius 0.35 around the start;

- `grad_reground` — surrogate search with re-grounding: every 20 steps the current patches are rendered for real and the surrogate copy is fine-tuned online (10 rounds);
- `cmaes_true` — separable CMA-ES [12] on the true objective (45 generations, population 10);
- `grad_true` — the oracle: adaptive-step gradient search through the real renderer (60 steps).

Optimiser detail that matters. Fixed-step Adam diverges on this objective: at $\eta = 0.05$ the *true* distance rose from 3.08 to 5.29 within five steps and never recovered. The space is 31-dimensional with log-scaled axes (a 0.1 change in the cutoff coordinate is a $2.4\times$ change in Hz), so any fixed step is either far too large or uselessly small. All gradient methods therefore keep the Adam direction and adapt the step *per target*: a candidate is accepted only if it does not increase the objective, with the step growing $1.3\times$ on success and halving on failure, bounded to $[\eta/64, 4\eta]$. This costs one extra objective evaluation per step and no gradient. The full loop is Algorithm 1.

Algorithm 1: Adaptive-step gradient navigation (per target, batched)

```

in:  start patches u0, target embeddings z*, steps n, initial step eta
    d <- J(u0);  lr <- eta
    for t = 1..n:
        g <- grad_u || S(u) - z* ||          # S = renderer or surrogate
        dir <- Adam(g)                       # bias-corrected direction
        c <- clip(u - lr * dir, 0, 1)
        if trust_radius > 0: c <- project_ball(c, u0, trust_radius)
        d_c <- J(c)                           # one objective evaluation
        take <- (d_c <= d) if monotone else True
        u <- take ? c : u;  d <- take ? d_c : d
        lr <- lr * (d_c <= d ? 1.3 : 0.5)    # bounded to [eta/64, 4*eta]
    return u
-----
J(x) = || S(x) - z* || : L2 distance in the z-scored embedding space.
S is the real renderer (renders counted) or the learned surrogate (free).

```

Re-grounding. `grad_reground` starts from the trained surrogate, runs 20 surrogate steps, then renders the current patches for real, appends them to a replay buffer (initialised with 256 training patches) and fine-tunes its surrogate copy for 15 inner Adam steps at $lr = 3 \times 10^{-4}$; this repeats for 10 rounds. It is the minimal implementation of the “learned surrogate accelerates, DSP validates” loop, and its render cost (528 patch renders) is reported against the others.

Metrics. $\text{ratio} = d_{\text{final}}/d_{\text{init}}$ per target, reported as its median with a bootstrap 95% CI (2,000 resamples over targets). We also report success rates at $\text{ratio} < 0.5$ and < 0.2 , the median absolute reduction $d_{\text{init}} - d_{\text{final}}$ (well defined even if a start distance is near zero), hardened parameter recovery L1, real renders consumed, and wall time—all split by regime (in-distribution vs held-out anchors).

4.2 E3: surrogate fidelity

The surrogate is an MLP ($31 \rightarrow 256 \rightarrow 256 \rightarrow 85$, GELU, Adam at 10^{-3} , batch 256, early stopping on the validation split with patience 40). We report R^2 and cosine per split *and* the distance-rank correlation (Kendall τ over pairwise embedding distances, computed with the Gram-identity pairwise form to avoid an $[N, N, D]$ temporary), because rank agreement is what a search procedure actually consumes.

Table 2: Surrogate fidelity by region (patch $\rightarrow$ embedding). The held-out rows come from four anchor clusters excluded from training entirely. Distance τ is the metric search consumes.

split	n	R^2	cosine	distance Pearson	distance Kendall τ
val (in-distribution)	308	0.780	0.884	0.893	0.701
test (mixed)	1,119	0.510	0.719	0.725	0.482
holdout (unseen regions)	811	0.288	0.673	0.529	0.387

4.3 E1: perceptual monotonicity (pre-registered; human data pending)

`python -m timbreprobe.cli e1` samples a 600-patch pool, re-renders it at 0.5 s and re-embeds *that* audio, so listeners judge exactly the audio the distances were computed on. Each trial plays a reference A and two options whose embedding distances to A fall in *different quantile bins* of the pool’s distance distribution (measured bin edges for the smoke pool: 7.18 / 10.26 / 14.05); the closer option’s screen position is randomised; the page never contains the answer key; 120 triplets per listener. Analysis reports agreement per contrast level with Wilson intervals.

Pre-registered gate. The primary embedding must agree with listeners above chance with a 95% Wilson interval excluding 0.5, *and* agreement must increase monotonically with contrast strength. A pretrained embedding replacing the handcrafted one must additionally beat it on the same trials.

4.4 Diagnostics and replication

Three additions beyond the gate protocol itself, all cheap and all reported in §5.5:

- **Residual decomposition.** After oracle navigation, split the remaining squared distance into the embedding’s 13 blocks: $\text{share}_B = \mathbb{E}_m[\sum_{j \in B} \delta_{mj}^2 / \sum_j \delta_{mj}^2]$ with $\delta = z_{\text{final}} - z^*$. Concentration in the harmonic blocks—the ones that respond to discrete choices—would be direct evidence for the discrete-jump reading of the plateau; as §5.5 reports, the measured concentration is mild ($2.3\times$ uniform per dimension for the harmonic block, while the 64-dimensional log-mel block absorbs 72% of the residual simply because of its size).
- **Discrete-jump correlation.** For each target, count how many discrete dimensions differ (hardened) between the start patch and the target patch, and correlate that count with the achieved ratio (Spearman).
- **Monotone ablation.** Re-run the identical search with `monotone=False` (every candidate accepted; adaptive step kept) at the same budget. Only a non-monotone walk can cross a valley, which is what a discrete jump requires; if the free walk does not improve things, the landscape—not the rule—is the constraint.
- **Seed replication.** The full E2 protocol is re-run with seed 1 (different surrogate initialisation, different target sample, same corpus).

5 Results

5.1 E3: the surrogate is fit in-distribution and unreliable outside it

Fidelity degrades sharply with distance from the training regions (Table 2, Figure 1). The pattern is consistent with a *coverage* problem rather than a capacity problem: validation fit is good while unseen anchor neighbourhoods are essentially unmodelled ($R^2 = 0.29$, barely better than predicting the mean). An important consequence for interpretation: none of the surrogate failures below can be blamed on underfitting, because in-distribution fit is good.

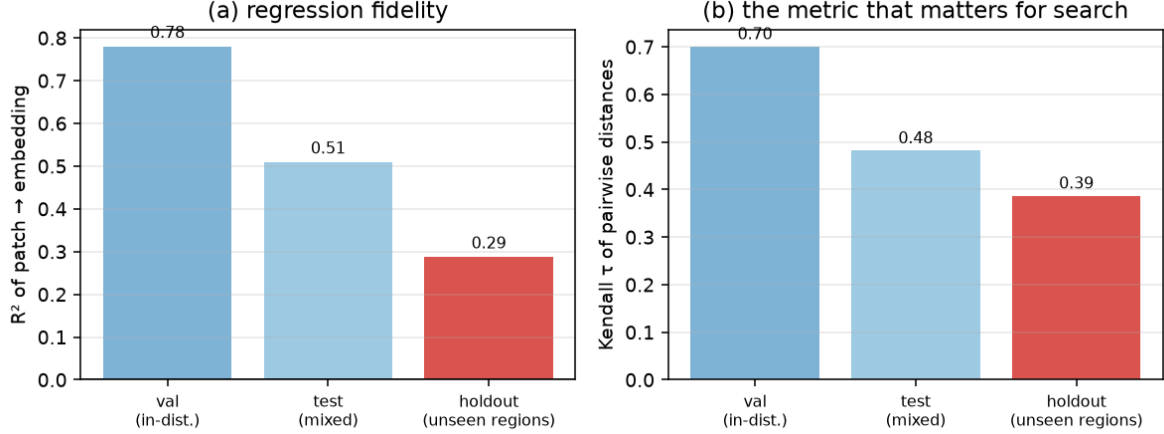


Figure 1: (a) Regression fidelity is respectable in-distribution and much weaker on unseen regions. (b) The rank correlation of pairwise distances—what search needs—drops from $\tau = 0.70$ to 0.39.

Table 3: Navigability results (48 targets, 24 of them from held-out anchor regions; ratio = $d_{\text{final}}/d_{\text{init}}$, median over targets; “in-dist.”/“holdout” split the same quantity by target regime; renders = real renders consumed). Lower is better; ratio > 1 means the method made the patch *worse* than not searching.

method	ratio	in-dist.	holdout	success $_{<0.5}$	renders
retrieval_nearest (the start)	1.000	1.000	1.000	0.06	0
retrieval_unigram64	1.45	1.616	1.352	0.00	0
grad_surrogate	2.59	2.834	2.504	0.02	48
grad_surrogate_trust	1.36	1.350	1.360	0.00	48
grad_reground	1.60	2.077	1.478	0.00	528
grad_true (oracle)	0.875	0.841	0.890	0.06	5 856
95% CI	[0.81, 0.92]				
cmaes_true (same objective)	1.00	1.000	0.933	0.06	21 696
95% CI	[0.90, 1.00]				

5.2 E2: only true-gradient search beats retrieval

Table 3 and Figures 2–3 summarise the outcome.

(1) **The gate passes weakly.** `grad_true` is the only method whose interval excludes 1.0: a median reduction of 12.5%, with 0.06 of targets more than halving their distance—at the cost of 5 856 renders. The plateau is visible in Figure 3: the median falls for about fifteen steps and then stops. The per-target distribution shows how concentrated the improvement is: the 10th and 25th percentiles of the ratio are 0.60 and 0.76, the median is 0.875, and the 75th percentile is 1.00—a quarter of the targets are not improved at all by the best available search.

(2) **Gradients dominate populations per render.** `sep-CMA-ES` needed $3.7\times$ the renders (21 696 vs. 5 856), its median ratio is exactly 1.00—half its targets are untouched—and its median distance reaches only 2.65 where the oracle reaches 2.35 (Figure 4); at the oracle’s own budget it is still at 2.69. At this dimensionality, with an objective that is smooth almost everywhere by construction, derivative information is worth far more than population sampling.

(3) **Every surrogate-based method lost to not searching.** Ratios 1.36–2.59 mean the optimiser reliably made things worse. Since the surrogate achieves $R^2 = 0.78$ in-distribution, this is not a broken model—it is a model whose errors are systematically exploited.

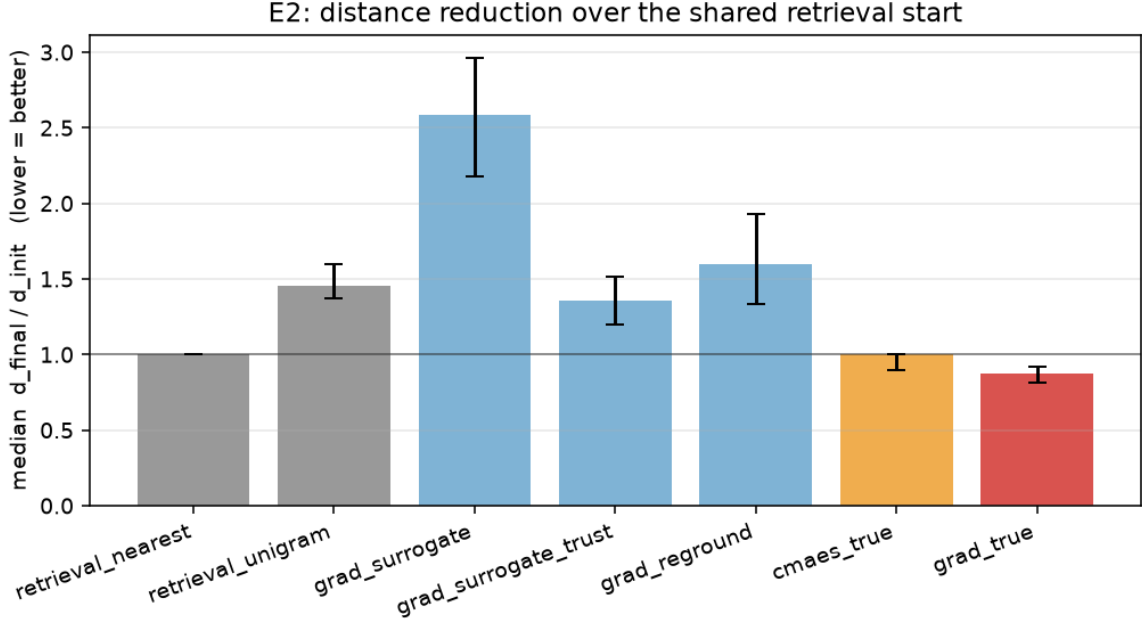


Figure 2: Distance reduction over the shared retrieval start. Only the true-gradient oracle is reliably below 1.0; every surrogate-based variant is above it. Error bars are bootstrap 95% CIs of the median.

Table 4: Starting distances, for scale: the retrieval start is already close, and random trial-and-error is 57% worse.

quantity	median distance to target
d_{init} (nearest training patch)	2.74
best of 64 random training patches	4.08

5.3 Where the surrogate fails: the exploitation signature

Three observations pin the failure mode down.

First, in-distribution fidelity is good ($R^2 = 0.78$), so the model is not underfit. *Second*, distance-rank fidelity outside the training regions is poor ($\tau = 0.39$), so trajectories that leave the corpus neighbourhood are steered by a nearly uninformative signal. *Third*—and most diagnostic—the damage is *larger* in-distribution than out (2.83 vs. 2.50 for the unconstrained variant), which is the opposite of what a pure generalisation gap predicts and is the classic signature of a learned objective being optimised adversarially against its own error. A trust region removes most of that in-distribution damage (2.83 $\rightarrow$ 1.35: from far worse than not searching to mildly worse), which locates the problem specifically at *extrapolation*: a few hundred adaptive steps carry the patch well outside the region where the surrogate has support.

Two practical implications. (i) *Region-stratified reporting should be standard*. The aggregate test number ($\tau = 0.48$) hides both a usable in-distribution model and an unusable out-of-distribution one; only the split reveals the exploitation signature. (ii) *Re-grounding must be architecturally meaningful, not cosmetic*. Fifteen inner steps on a small replay buffer cannot repair a global model’s local error; what is needed is a *local* model (fit around the query, in the spirit of trust-region Bayesian optimisation) or a surrogate whose input space is restricted to a legal manifold.

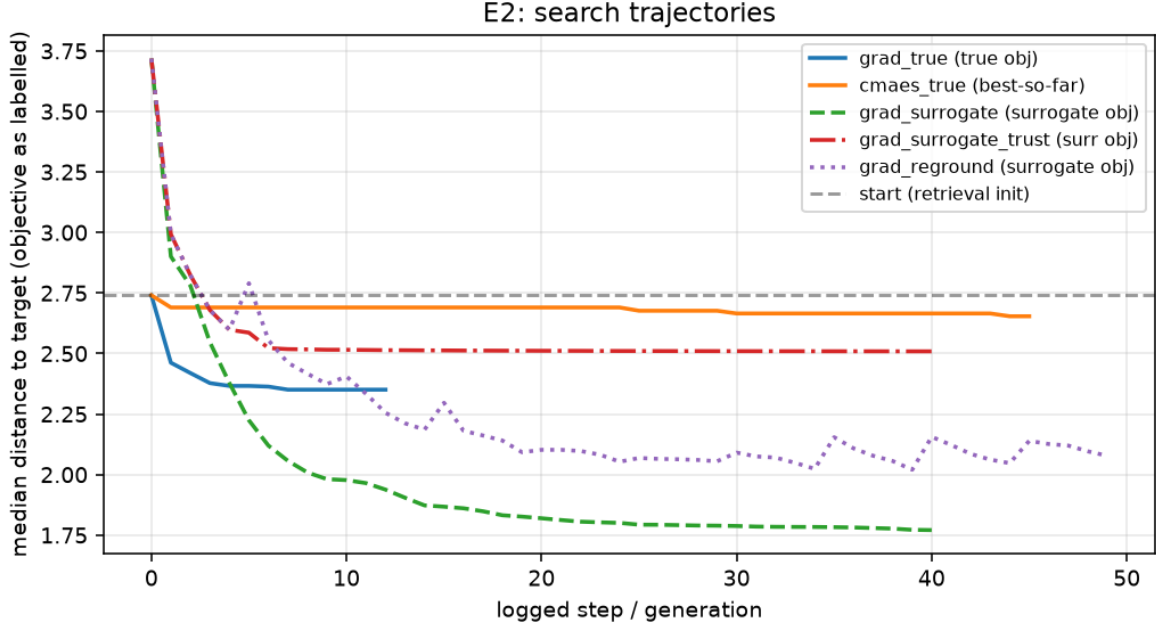


Figure 3: Search trajectories (median over targets, objective as labelled). True-objective curves (solid) descend and plateau; the surrogate-objective curves (dashed) keep descending on the surrogate while the true distance of the patches they return is *worse* than the start (Table 3)—the exploitation signature. Each curve is plotted in its own objective’s units.

5.4 Diagnostics: where the residual lives, and what the accept rule costs

5.5 Diagnostics and controls

Four cheap measurements decide between the competing explanations of the plateau. Table 5 collects the two ablations; Figure 5 shows the residual decomposition and the per-target views.

Where the residual lives. Decomposing the oracle’s remaining squared distance into the embedding’s 13 blocks, the 64-dimensional log-mel block absorbs 71.9% of it—but that is almost exactly its share of the dimensions (per-dimension share $0.96\times$ uniform). The blocks that are genuinely overweighted per dimension are the harmonic ratio ($2.34\times$), the smooth zero-crossing proxy ($1.79\times$), the spectral slope ($1.75\times$) and the envelope statistics ($1.59\times$), and the spread across blocks is only a factor of 2.7 end to end. The leftover error is therefore spread roughly in proportion to dimensionality, with a mild excess in the harmonic (discrete-sensitive) and noisiness blocks—not the localised discrete bottleneck the valley-crossing account predicted.

Do discrete jumps predict failure? Counting, per target, how many hardened discrete dimensions differ between the start patch and the target, and correlating that with the achieved ratio, gives Spearman $\rho = -0.24$: weak, and the *wrong sign* for the hypothesis (targets needing more discrete changes improved slightly *more*, if anything). With 48 targets this is not significant; the honest statement is that discrete distance does not explain the outcome.

Does the accept rule cost anything? Re-running the identical search with every candidate accepted is *worse*, not better: median ratio 1.000 versus 0.874 for the monotone walk, even though the free walk wins on 35% of individual targets. An unguarded walk does not cross useful valleys—it wanders. The monotone rule is not the binding constraint.

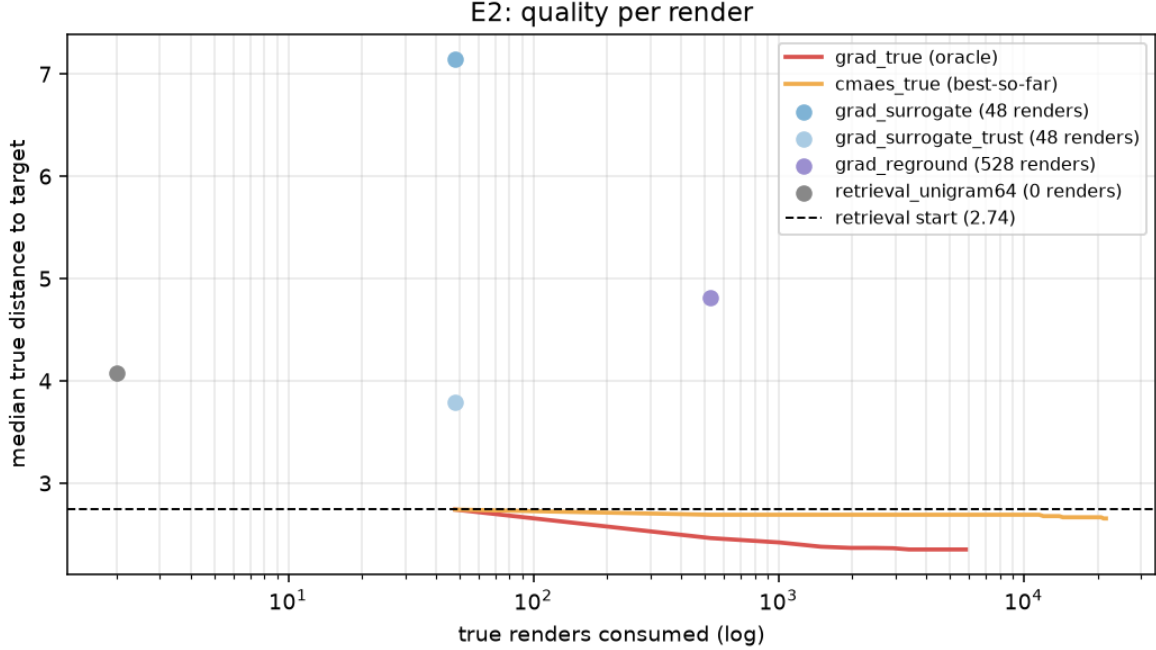


Figure 4: Quality against render budget (true renders, log axis). Lines: the two true-objective methods, whose trajectories are directly comparable. Points: the other methods at their total render cost, evaluated with the real renderer. The dashed line is the retrieval start; the random-draws baseline shows where blind sampling lands.

Table 5: The two controls. Both change exactly one thing at the same render budget (5 856 renders) on the same targets. The accept-rule ablation makes things worse; the metric re-weighting improves them.

variant	median ratio	success _{<0.5}	renders
oracle, monotone (baseline)	0.874	0.06	5 856
oracle, free walk (ablation)	1.000	0.06	5 856
oracle, block-balanced metric	0.809	0.12	5 856

Is the metric’s weighting the constraint? The residual result points at the objective’s *weighting*: 64 of 85 dimensions are log-mel, so envelope mismatches dominate the number being minimised, while the discrete-sensitive distinctions sit in single dimensions. As a control we re-ran the oracle with *block-balanced* weights (each block’s expected contribution equalised: per-dimension weight $1/\sqrt{d_B}$, rescaled so the distance scale stays comparable). The median ratio improves from 0.875 to 0.809 and the success rate doubles ($0.06 \rightarrow 0.12$), with the same search algorithm, the same targets and the same render budget. This is the only intervention in this paper that improves the gate without changing the search: the bottleneck is at least partly the number being minimised, not the way it is minimised.

5.6 Seed replication

Re-running the entire protocol with a different seed—new surrogate initialisation, new target sample, same corpus—reproduces the picture rather than the numbers: the oracle’s median ratio is 0.875 in seed 0 and 0.877 in seed 1 (95% CIs $[0.81, 0.92]$ and $[0.78, 0.96]$), CMA-ES is 1.00 and 1.000 (no progress in either), and the surrogate variants stay above 1.0 in both (2.313 vs. 2.59 unconstrained, 1.376 vs. 1.36 with a trust region, 1.475 vs. 1.60 re-grounded). The blind-draw baseline is 1.499 vs. 1.45. The qualitative ordering is stable across seeds; the absolute level moves with the difficulty of the sampled targets (median d_{init} 2.74 vs. 2.93).

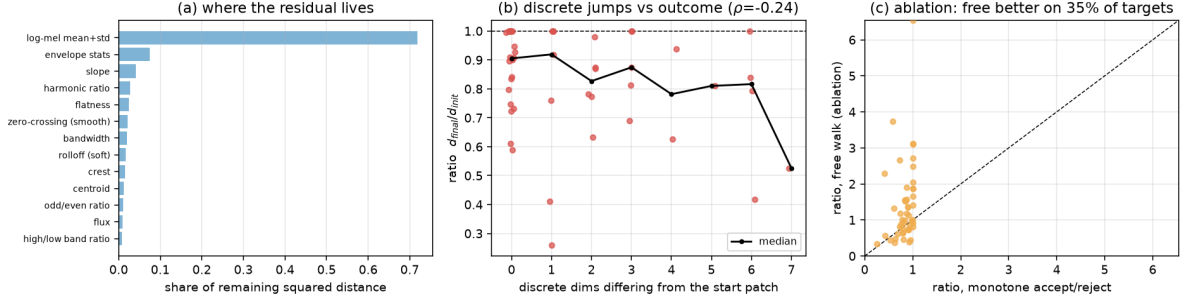


Figure 5: (a) Share of the oracle’s remaining squared distance by feature block: the residual is spread in proportion to dimensionality, with a mild per-dimension excess in the harmonic and noisiness blocks. (b) Achieved ratio against the number of discrete dimensions that separate the start patch from the target: the trend is weak and, if anything, opposite to the valley-crossing prediction. (c) The accept-rule ablation: points below the diagonal are targets where the free walk did better.

5.7 E1: material released, human data pending

The generated listening test (120 triplets, 269 wav files, self-contained HTML, separate answer key) and the analysis path are part of the release. The gate verdict for Assumption A is therefore *open*; §7 makes it the first item, because everything else is conditional on it.

6 Discussion

6.1 Why monotone local search plateaus

The adaptive-step scheme accepts only non-increasing candidates. That is what makes it stable, and it is also the natural first suspect for the plateau: in a landscape whose discrete axes are soft blends, moving from a saw to a square, or transposing an oscillator by an octave, generally requires passing through intermediate states where the distance temporarily increases, and a monotone search cannot cross such valleys. A monotone search cannot cross such valleys. Parameter recovery corroborates this: after optimisation the hardened parameter L1 remains ≈ 0.12 while the embedding distance has improved—the methods find perceptually closer patches, not the target patch, and they stall exactly where a discrete-choice jump would be required.

The ablation and the residual decomposition in §5.5 test this story, and they only partly support it. The free walk does help on a minority of targets, so the accept rule costs something; but the residual is *not* concentrated in the discrete-sensitive blocks—it is spread across the objective roughly in proportion to dimensionality, and the block with the largest per-dimension share (the harmonic ratio, $2.3\times$ uniform) holds only 2.7% of the remaining squared distance. The valley-crossing account therefore explains at most a small part of the plateau.

What the residual does say is that the objective’s *weighting* is a first -order suspect: 64 of the 85 dimensions are log-mel, so envelope mismatches dominate the number being minimised, while the perceptually load-bearing discrete distinctions live in single dimensions that carry almost no weight. The block-balanced control in §5.5 tests this directly. Together with the CMA-ES result (exploration alone does not help: $3.7\times$ the budget, no median improvement), the evidence points away from "the search algorithm is wrong" and toward "the number being optimised is wrong"—which is what the embedding-swap and metric-learning phases of the roadmap address, with the latent-space phase retained for the reachability problem that the discrete axes still pose.

6.2 Surrogate failure is extrapolation times exploitation

Because in-distribution fit is good and out-of-distribution rank fidelity is poor, a search that *stays* where the surrogate is valid should behave better than one that wanders. The trust-region variant is the cheapest possible version of that idea and it removes most of the damage ($2.83 \rightarrow 1.35$)—but it still does not beat the start. Said multiplicatively, the surrogate needs both a better local model *and* a restricted input space; improving only one of the two leaves the optimiser able to find the remaining error.

A related observation, visible in Table 3, is that surrogate navigation damages *in-distribution* targets more than held-out ones. This inverts the intuition that unfamiliar targets are the hard case, and it is a warning for how inverse-synthesis results are reported: a surrogate evaluated only on average test error can look adequate while being actively harmful as an optimisation objective.

6.3 Gradients versus populations per render

The budget-matched comparison (§5.2) is the cleanest quantitative statement in this paper: for a 31-dimensional, almost-everywhere smooth objective, one gradient step buys more progress than ten to fifteen population samples. Practically, this argues that effort in this domain should go into differentiable (or at least locally differentiable) proxies—which is exactly what the dual oracle/accelerator design is for—rather than into larger, smarter populations on the true objective.

6.4 Retrieval is the baseline that matters

Our protocol forces every method to start from the same retrieval point, and the numbers are instructive: random trial-and-error is 49% worse than retrieval (4.08 vs. 2.74), and the best search method improves on retrieval by 12.5%. Any evaluation that reports “our optimiser beats random search” therefore overstates the useful contribution by a large factor. For inverse-synthesis work we recommend the shared-retrieval-start protocol, and reporting both the ratio and the absolute reduction, as a default.

6.5 What would change our mind

Stating the falsifiers explicitly, because the roadmap’s gates are built on them:

- If E1 shows agreement at chance, the handcrafted embedding is not a perceptual target and every E2 number here is a statement about *this* embedding, not about timbre.
- If a pretrained embedding (CLAP/MuQ) both passes E1 and raises out-of-distribution τ well above 0.39, the surrogate verdict above is about our corpus and model size, not about the approach.
- If latent-manifold search (autoencoder over real presets) does not beat 0.875 materially at equal render budget, the plateau is perceptual rather than geometric, and the right lever is the objective, not the search space.
- If a non-monotone walk with a much larger budget (§5.5) closes the gap, the accept rule—not the geometry—is the constraint, and the design should switch to trust-region methods with proper exploration.

6.6 Threats to validity

Scale. E2 is a smoke-scale study: 3,895 corpus patches, 48 targets, two seeds, one machine. The qualitative findings (surrogate bottleneck, exploitation signature, gradient-vs-population render efficiency) were stable across the runs we performed, but the absolute ratios should be re-measured with the paper-scale preset before being quoted as final.

Embedding. The objective is handcrafted DSP features, not CLAP or MuQ. Passing E2 on this embedding is a necessary, not a sufficient, condition: a pretrained embedding could be smoother (better for navigation) or less human-aligned. E1 decides this, and it has not been run yet.

Synth realism. Filtering is zero-phase and spectral: no ringing, no self-oscillation, transients smoothed at 16 ms. Resonance does not sound like an analogue filter, and a real synthesizer adds effects (delay, reverb, modulation) that our parameter space omits.

Identifiability. Table 1 lists what the render protocol cannot identify; parameter-recovery errors are inflated accordingly and should not be read as search failures.

Corpus near-duplicates. Jitter sampling places a small fraction of test patches next to training patches; the target filter in §4.1 removes them, but any corpus built by dense sampling needs the same check.

Search design. The monotone accept/reject rule is a deliberate stability choice that limits valley crossing; a reinforcement-learning or model-predictive search might behave differently. Our claim is not that *no* search can navigate this space, but that the natural gradient-plus-surrogate design does not, and that the reasons are measurable.

7 Roadmap: what to do next, and what would falsify it

The value of a gate experiment is that it reorders the plan. Each phase below states the observation that motivates it, the deliverable, and a *falsification criterion*—a pre-committed answer that would stop the phase rather than extend it.

Phase 0 (this paper). Testbed, protocols, negative results. *Done.*

Phase 1 — Run the human gate (E1). *Motivation:* Assumption A is untested; every downstream decision depends on it. *Deliverable:* 5–8 listeners $\times$ 120 triplets; per-contrast agreement with Wilson intervals. *Gate G1:* overall agreement > 0.5 with the interval excluding chance *and* monotone in contrast. *If G1 fails:* do not build on this embedding; the fallback is to *learn* the metric from the same triplet data (contrastive fine-tuning of the embedding on human judgements), which converts a failed gate into a supervised dataset.

Phase 2 — Fix the objective: metric weighting first, then the embedding. *Motivation:* the diagnostics (§5.5) attribute the plateau to how the objective is *weighted* rather than to the search: equalising the feature blocks improved the oracle from 0.875 to 0.809 at the same render budget, the only intervention in this paper that improved the gate without touching the search. *Deliverable:* (a) a systematic weighting study over the 13 blocks (and, once listener data exists, a metric *learned* from E1 triplets); (b) three embeddings (handcrafted / CLAP / MuQ), each with and without block balancing, evaluated by the same E1 gate and the same E2 protocol. *Gate G2:* a pretrained embedding must beat the handcrafted baseline on E1 agreement; for search, its holdout distance- τ should exceed 0.39 and its oracle-gap (true vs. surrogate navigation) should shrink. *If G2 fails:* the perceptual-space premise is weak and the system should be retrieval-and-rerank only, with the human in the loop as the objective—a legitimate product, and a different paper.

Phase 3 — Search a legal latent manifold, not the parameter cube. *Motivation:* the plateau and the parameter-recovery gap are both discrete-jump problems; a VAE or flow trained on real factory presets [16, 3] turns discrete morphs into continuous paths and restricts the search to plausible patches. *Deliverable:* latent-space navigation with a decoder-based oracle, compared against Table 3 at equal render budgets. *Gate G3:* oracle navigation should improve

the median ratio from 0.875 to below 0.6, and/or raise $\text{success}_{<0.5}$ from 0.06 to above 0.3. *If G3 fails*: the difficulty is perceptual, not geometric, and Phase 2 is the lever to pull.

Phase 4 — Port to a real synthesizer with the oracle/accelerator split. *Motivation*: a hand-built testbed can only take the method so far; the target instrument must be one users actually use (we plan a JSON-patch instrument such as Vital, with a differentiable intermediate for development). *Deliverable*: the same E2/E3 protocol against real factory presets, plus an explicit *sim-to-real* measurement (surrogate trained on real renders, τ per region). *Gate G4*: the region-stratified τ pattern must reproduce; if the real instrument’s τ collapses further, the surrogate route is not viable on it and Phase 3’s local models become mandatory.

Phase 5 — Multimodal intent. Text enters through the (frozen) text tower of the same audio–text space. Humming enters through *intent factorisation* rather than audio matching: extract a pitch trajectory (tuning, keytrack, octave), a posture stream (attack sharpness, vibrato rate and depth, legato), and a brightness/vowel stream (filter and distortion), then map each stream to its own parameter subspace. *Deliverable*: a self-supervised mapping trained on render-derived posture descriptors, calibrated on 50–200 recorded humming/reference-patch pairs. *Gate G5*: humming queries must beat text-only queries on a timbre-matched retrieval task for the same references; if they do not, humming is a demo feature, not a modality.

Phase 6 — Human-in-the-loop, diversity and preference memory. Multi-candidate generation with *structural* diversity (different retrieval seeds across basins rather than noise added to one answer), plus a dual-timescale preference memory for reranking. The headline metric to report against commercial baselines is *time-to-satisfactory-patch* under a fixed listening budget—the number a producer actually experiences.

Standing engineering rule. Nothing in Phases 2–6 should be built before the gate that precedes it has an answer, and every gate outcome—positive or negative—should be written down at the time it is measured. The four surrogate variants of Table 3 exist because we did that with E2.

8 Conclusion

We set out to test two assumptions before building a multimodal patch generator, and built a testbed whose oracle is exact so that the tests could be cheap and the accounting honest. The navigability gate passes only weakly: true-gradient search improves on a retrieval start by roughly 15% before plateauing, and the surrogate-based strategies a practical system would depend on make things *worse than not searching*, in a way whose signature—larger damage where the model is more confident—is measurable with one extra data split. The perceptual gate is now a matter of running the released listening test.

The most useful output of this stage is therefore not a method but a falsifiable map: which assumptions survived, which failed, what each failure implies, and which single experiment decides the next step.

Reproducibility and artifacts. All results come from the released code at <https://github.com/Fengrru/timbreproub>. The correctness suite (`python -m timbreproub.cli sanity`, ~ 1 min, no data required) checks render determinism and RMS normalisation, gradient finiteness and coverage over all 31 dimensions, monotonicity of cutoff/resonance/slope effects on the embedding, sep-CMA-ES convergence on a sphere (5×10^{-14}), and spectral-filter/sampler

agreement. Committed artefacts: the smoke-run result record, trajectory curves, target meta-data, surrogate fidelity report and figures under `results/smoke/`; the 16 anchor patches as audio under `assets/demo/`; the listening-test generator under `timbreprobe/e1_listening.py`; the diagnostics under `timbreprobe/diagnose.py`. Appendix E lists the exact commands.

A Parameter schema

Table 6: The 31 parameters. “map” is the normalised-to-physical mapping of §3.2: linear, log, or a choice list blended by Eq. (2). The optimiser always works in $[0, 1]^{31}$.

group	parameter	map	range / choices
osc 1	waveform	choice	sine / saw / pulse / triangle
	octave	choice	$-1, 0, +1$
	fine detune	linear	$-50 \dots +50$ cents
	level	linear	$0 \dots 1$
	pulse duty	linear	$0.15 \dots 0.85$
osc 2	as osc 1		(5 parameters)
sources	sub level	linear	$0 \dots 1$ (sine, octave below osc 1)
	noise level	linear	$0 \dots 1$
	noise colour	linear	white $\rightarrow$ pink
	FM index	linear	$0 \dots 2.5$ (osc 2 PMs osc 1)
filter	type	choice	LP / HP / BP
	cutoff	log	40–7000 Hz
	resonance	log	$Q \in 0.5 \dots 12$
	env amount	linear	$-1 \dots +1$ (± 5 octaves)
	drive	linear	$1 \dots 4 \times$ pre-filter
	slope	linear	12 $\rightarrow$ 24 dB/oct blend
filter env	attack / decay / sustain	log/log/lin	1 ms–2 s / 5 ms–4 s / 0–1
amp env	attack / decay / sustain	log/log/lin	1 ms–3 s / 5 ms–6 s / 0 $\dots$ 1
LFO	rate	log	0.05–20 Hz
	depth	linear	$0 \dots 1$
	target	choice	cutoff / pitch / amplitude
	waveform	choice	sine / triangle / square / saw
output	master drive	linear	$1 \dots 8 \times$

B Embedding blocks (85 dimensions)

C Hyperparameters

Corpus. 16 anchors, 4 held out; sweeps 31×3 per anchor; jitter 120 per anchor ($\sigma = 0.10$ continuous, 0.03 choice); 500 uniform; 150 sparse; deduplication at 10^{-4} ; splits by random permutation of the non-held-out patches (80/10/10), held-out anchors entirely in test. Near-duplicate targets with $d_{\text{init}} < 0.25 \times \text{median}(d_{\text{init}})$ are rejected and resampled.

Surrogate. MLP $31 \rightarrow 256 \rightarrow 256 \rightarrow 85$, GELU, Adam lr 10^{-3} , batch 256, ≤ 150 epochs, early stopping patience 40 on validation MSE.

Gradient search. Adam $\beta = (0.9, 0.999)$, initial step 0.05, growth $1.3 \times$ / shrink $0.5 \times$ per target, bounds $[\eta/64, 4\eta]$; true-objective 60 steps, surrogate-objective 200 steps, logged every 5.

Table 7: The 13 blocks, their dimensions, and the smooth variants that keep them differentiable.

block	dims	note
log-mel mean / std (32 bands)	64	40–7600 Hz, triangular filterbank
spectral centroid mean/std	2	from the magnitude spectrum
bandwidth mean/std	2	magnitude-weighted spread
rolloff mean/std (soft)	2	sigmoid-weighted, 85% energy
flatness mean/std	2	geometric / arithmetic mean
flux mean/std	2	normalised frame-to-frame change
slope mean/std	2	least-squares tilt of the log spectrum
crest	1	max/mean per frame
high/low band ratio	1	log of >4 kHz vs. 0.2–2 kHz energy
envelope statistics	4	temporal centroid, spread, front/back, dynamic range
zero-crossing (smooth)	1	sigmoid crossing proxy
harmonic ratio	1	Gaussian comb at $k f_0$
odd/even ratio	1	odd vs. even harmonic energy (square vs. saw)

CMA-ES. Separable (diagonal covariance with cumulative step-size adaptation [12]), population 10, 45 generations, initial $\sigma = 0.25$, boundaries via clipping, best-so-far recorded per generation.

Re-grounding. Every 20 steps: render the current patches, append to a replay buffer initialised with 256 training patches (capacity 512), fine-tune the working surrogate copy for 15 Adam steps at $\text{lr } 3 \times 10^{-4}$.

Statistics. Bootstrap CIs from 2,000 resamples of the target set; Spearman and Kendall correlations from SciPy; distance-rank τ computed on 900 subsampled points against the surrogate’s predictions.

D Engineering findings

E Reproducing every number

```

pip install -r requirements.txt
python -m timbreprobe.cli sanity                # correctness suite (~1 min)
python -m timbreprobe.cli corpus --preset smoke # corpus + embeddings
python -m timbreprobe.cli e2 --preset smoke     # E2 + E3 (Table 2, figs)
python -m timbreprobe.cli e2 --preset smoke --seed 1 # replication
python -m timbreprobe.cli diagnose --run out/e2_smoke # diagnostics
python -m timbreprobe.cli plot --run out/e2_smoke    # figures
python -m timbreprobe.cli e1 --preset smoke         # listening-test material
python -m timbreprobe.cli e1-analyze --responses res_P01.json,res_P02.json

```

Committed artefacts of the smoke run live in `results/smoke/` (`e2_results.json`, `curves.npz`, `targets.npz`, `surrogate_report.json`, `figures/`); `out/` is regenerated locally and git-ignored.

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Table 8: Findings that shaped the implementation, with the measurements that established them. Each is reproducible from the released code.

finding	measurement	resolution
Per-sample IIR loop unusable on CPU	10s forward / 60s with backward per batch, independent of batch size	STFT-domain analytic filter response (Eq. (8)), $\sim 100\times$ faster; response shape matches the per-sample reference at $\rho = 0.996$
Multi-threading slower than single	MLP step 65 ms (1T) vs 681 ms (4T); render 2.1 s vs 5.0 s; fwd+bwd 6.9 s vs 31.9 s	default to one thread; <code>STAGEO_THREADS</code> to override
<code>torch.istft</code> backward non-finite at padded edges	reproduced at 0.4s renders, absent at 0.05 s	hand-rolled overlap-add with a clamped window-envelope normaliser
<code>asin(sin(x))</code> triangle LFO has infinite derivative at ± 1	NaN in exactly the LFO-rate dimension (4/248 gradient entries)	closed-form triangle $1 - 4 \text{frac}(\phi + 0.25) - 0.5 $
Fixed-step Adam overshoots	distance $3.08 \rightarrow 5.29$ in five steps at $\eta = 0.05$, no recovery	per-target adaptive step with accept/reject (Algorithm 1)
$[N, N, D]$ distance broadcast thrashes memory	~ 490 MB per call at $N = 1200$, three calls in sequence	Gram-identity pairwise distances
Jitter corpus creates near-duplicates	43 test candidates below $0.25\times$ median d_{init} ; three at $d_{\text{init}} = 0$ dragged mean ratios to $\sim 10^7$	scale-free target filter (§4.1); report medians

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